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On-Device Autonomous Software Engineering: In-Memory Code Property Graph Reachability Slicing under Edge VRAM Constraints

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Abstract:	State-of-the-art software engineering (SWE) agents rely on monolithic, cloud-hosted large language models with massive context windows (128k+ tokens) running on datacenter clusters. Deploying SWE agents on consumer edge hardware (< 8 GB VRAM) triggers catastrophic Out-of-Memory (OOM) crashes due to the Key-Value (KV) cache memory wall, while multi-turn loops induce exploration drift. We present NeuroCode, a lightweight systems architecture that offloads structural codebase comprehension to the idle Host CPU and DRAM. NeuroCode parses target repositories into an in-memory Code Property Multigraph (CPG) using Tree-sitter in 4.28 ms. Given a bug report, it extracts a bidirectional reachability closure and knapsack-packs critical context in 0.62 ms, strictly bounding prompts below 1,500 tokens. Transferring this payload across PCIe takes only 24.7 µs, enabling single-turn patch synthesis on edge GPUs without multi-turn tool loops. Evaluated on 292 tasks from SWE-bench Lite, NeuroCode achieves a 97.1% prompt token reduction (1,314 vs. 44,687 tokens), cuts developer inference costs by 97.1% ($0.0039 vs. $0.1341), improves fault localization Recall@3 by 5.5x over BM25 and RepoMap (17.1% vs. 3.1%), achieves a 41.4% valid patch application rate, and delivers 9.5% single-turn unit-test resolution compared to 0.0% (fatal OOM) for un-sliced baselines. Physical deployment on an NVIDIA GeForce RTX 3050 Laptop GPU (3.67 GB VRAM) confirms stable patch synthesis consuming only 961.2 MB of active VRAM at 15.8 tokens/second, operating entirely within RAM tmpfs without NVMe drive wear.

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November 10, 2026

To: The Editor-in-Chief
Journal of Parallel and Distributed Computing (JPDC)
Elsevier B.V.

Subject: Submission of Research Article titled *"On-Device Autonomous Software Engineering: In-Memory Code Property Graph Reachability Slicing under Edge VRAM Constraints"*
(Manuscript Transferred via Elsevier Transfer Service from Journal of Systems and Software)

Dear Prof. Sartaj Sahni and Esteemed Editorial Board Members,

I am pleased to submit our original research manuscript titled **"On-Device Autonomous Software Engineering: In-Memory Code Property Graph Reachability Slicing under Edge VRAM Constraints"** for consideration for publication as a regular research paper in the *Journal of Parallel and Distributed Computing* (JPDC) under the **Traditional (Subscription-based, $0 APC)** publishing model.

Context of Transfer & Editorial Recommendation

This manuscript was initially submitted to the *Journal of Systems and Software* (Ref: JSSOFTWARE-D-26-03025). The Editor, Dr. Alexander Chatzigeorgiou, recommended that the paper's core contributions are centrally positioned in **Edge Computing, Heterogeneous Architectures, and Parallel AI Systems**, advising a transfer to a leading venue specialized in distributed systems and accelerator architectures. Given JPDC's high standing in parallel algorithms, edge/accelerator co-design, and memory-constrained computing, we are excited to submit this work to your esteemed journal.

Summary of Contributions and Fit for JPDC

State-of-the-art autonomous software engineering agents rely on datacenter-scale GPU clusters to ingest 100k+ token code repositories. Deploying such systems on resource-constrained edge accelerators (< 8 GB VRAM) triggers immediate Out-of-Memory (OOM) crashes due to the quadratic Key-Value (KV) cache memory wall.

Our paper presents **NeuroCode**, a heterogeneous hardware-software co-designed architecture that resolves this fundamental memory wall: 1. **Asymmetric Host-Device Partitioning**: We offload structural codebase comprehension and reachability analysis entirely to idle Host CPU and DRAM, preserving precious accelerator VRAM strictly for autoregressive neural generation. 2. **Sub-Millisecond Graph Reachability**: Utilizing an in-memory Code Property Multigraph (CPG), NeuroCode extracts bidirectional reachability closures and knapsack-packs critical execution subgraphs in **0.62 ms**, bounding generation context strictly below 1,500 tokens. 3. **Micro-Benchmarked PCIe Communication**: The sliced payload transfers across the PCIe bus in only **24.7 µs**, eliminating multi-turn tool-calling exploration drift and enabling single-turn patch synthesis. 4. **Physical Edge Accelerator Validation**: Evaluated on 292 tasks from SWE-bench Lite and physically deployed on an NVIDIA GeForce RTX 3050 Laptop GPU (3.67 GB usable VRAM), NeuroCode

achieves a **97.1% token reduction**, reduces inference VRAM to **961.2 MB**, and delivers a **9.5% single-turn resolution rate** compared to **0.0% (fatal CUDA OOM crash)** for standard agent baselines.

Declarations

- **Originality**: This manuscript represents original research and is not under consideration for publication elsewhere. A preliminary 4-page tool demonstration describing the VS Code extension and MCP server appears in the Tool Demonstration Track of IEEE/ACM ICPC 2027; this journal article substantially extends that work with full formal graph formulations, multi-model VRAM scaling theorems, physical PCIe microbenchmarks, and comprehensive empirical evaluations.
- **Publishing Model**: We confirm that we select the **Traditional Subscription model ($0.00 APC)**.
- **Conflicts of Interest**: The author declares no conflicts of interest.
- **Reproducibility**: All source code, dataset artifacts, pre-computed graph slices, and reproduction scripts are publicly hosted with frozen release tags.

Thank you very much for your time, consideration, and editorial guidance.

Sincerely,

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Research Highlights

On-Device Autonomous Software Engineering: In-Memory Code Property Graph Reachability Slicing under Edge VRAM Constraints

- Heterogeneous CPU-GPU co-design overcomes the KV-cache memory wall on edge devices.
- In-memory Code Property Graph reachability closure extracts context in 0.62 ms.
- Sub-millisecond PCIe transfer strictly bounds prompt size below 1,500 tokens.
- Physical RTX 3050 deployment confirms 961 MB active VRAM with zero disk wear.
- Achieves 9.5% single-turn SWE-bench Lite resolution vs. 0.0% OOM baseline failures.

Manuscript Click here to view linked References

On-Device Autonomous Software Engineering: In-Memory Code Property Graph Reachability Slicing under Edge VRAM Constraints

Deepak R., [1]

Abstract

State-of-the-art software engineering (SWE) agents rely on monolithic, cloud-hosted large language models (LLMs) with massive context windows (128k+ tokens) running on dense datacenter clusters. Deploying SWE agents on consumer edge hardware (< 8 GB VRAM) introduces a catastrophic failure mode: the *Key-Value (KV) Cache Memory Wall*. Storing attention KV tensors for full repository dumps requires up to 14.2 GB of VRAM, instantly triggering Out-of-Memory (OOM) kernel panics. Concurrently, multi-turn tool-calling loops cause compact edge models to enter repetitive exploration drift.

We present **NeuroCode**, a lightweight systems architecture that shifts structural codebase reasoning to the idle Host CPU and system DRAM. NeuroCode parses target repositories into an in-memory Code Property Multigraph (CPG) using incremental Tree-sitter parsers in 4.28 ms. Given a bug report, it extracts the bidirectional reachability closure and knapsack-packs it on the Host CPU in only **0.62 ms**, strictly bounding the prompt below 1,500 tokens. Transferring this minimal payload across the PCIe bus takes only 24.7 μs, allowing the edge GPU to synthesize patches in a single generation turn without multi-turn tool loops.

Evaluated across 292 real-world tasks from SWE-bench Lite, NeuroCode achieves a **97.1% prompt token reduction** (1,314 vs. 44,687 tokens), cuts developer inference cost by **97.1%** ($0.0039 vs. $0.1341 per bug attempt), improves fault localization Recall@3 by **5.5**× over BM25 and RepoMap (17.1% vs. 3.1%), achieves a **41.4% (121/292) valid patch application rate**, and delivers a **9.5% (28/292) single-turn unit-test resolution rate** compared to 0.0% (Fatal CUDA OOM crash) for un-sliced baselines. Physical deployment on an NVIDIA GeForce RTX 3050 Laptop GPU (3.67 GB VRAM) demonstrates stable on-device patch synthesis consuming only **961.2 MB of active inference VRAM** at 15.8 tokens/second, operating entirely within tmpfs without NVMe solid-state drive wear.

Deepak R.: On-Device Autonomous Software Engineering under Edge VRAM Constraints

Autonomous software engineering, Large Language Models, Code Property Multigraph, edge computing, program repair, Key-Value cache, hardware-software co-design.

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[1] Deepak R. is with the Department of Robotics and Automation Engineering, Dhaanish Ahmed College of Engineering, Anna University, Chennai 601301, India (e-mail: wssedd18@gmail.com). A preliminary 4-page tool demonstration of the VS Code extension and MCP server appears in the Proceedings of the 35th IEEE/ACM International Conference on Program Comprehension (ICPC 2027) Tool Demonstration Track (R. 2027). This journal submission extends that work with full formal graph formulations, multi-model VRAM scaling proofs, physical PCIe transfer microbenchmarks, and empirical ablation over 292 SWE-bench Lite defects.

Introduction

software engineering (SWE) has emerged as one of the most commercially significant applications of foundation models. Benchmarks such as SWE-bench (Jimenez et al. 2024) evaluate whether autonomous agents can ingest natural language bug descriptions, localize faulty code symbols across thousands of files, and synthesize syntactically valid unified diff patches that pass regression test suites.

Modern industrial agents (e.g., SWE-agent (Yang et al. 2024), Agentless (Xia et al. 2024)) predominantly operate on high-density cloud infrastructure powered by clusters of NVIDIA H100 or A100 GPUs. These systems ingest expansive repository contexts—often exceeding 30,000 to 100,000 tokens—by exploiting 128k+ token context windows and distributed Key-Value (KV) cache paging (Kwon et al. 2023).

However, deploying SWE agents on consumer edge devices (e.g., standard developer laptops, edge workstations with 4 GB to 8 GB VRAM) reveals three fundamental systems bottlenecks:

1. **The KV-Cache Memory Wall**: Autoregressive transformer inference requires caching Key and Value activation vectors for every prior token across all attention heads and layers. When feeding an average repository context of 35,000 tokens to an 8B/9B model (e.g., Gemma-2-9B (Gemma Team et al. 2024)), the KV cache alone demands 12.0 GB to 14.2 GB of GPU memory, instantly exceeding the physical memory boundary of consumer accelerators.

2. **Exploration Drift in Multi-Turn Tool Loops**: Existing agents rely on ReAct-style iterative command loops (issuing shell commands like `grep`, `find`, and `cat`). While 70B+ parameter cloud models maintain state across 15–30 interaction turns, compact 1.5B–9B edge models suffer from severe exploration drift: they enter degenerate infinite loops, repeat failing search commands, and exhaust their token budget within 3–5 turns.

3. **Lexical Blindness**: Systems relying on BM25 lexical search (Xia et al. 2024) retrieve files containing exact query keywords. In real-world software architectures, callers and downstream dependencies rarely share lexical overlap with the bug report, causing lexical retrievers to miss root-cause definitions.

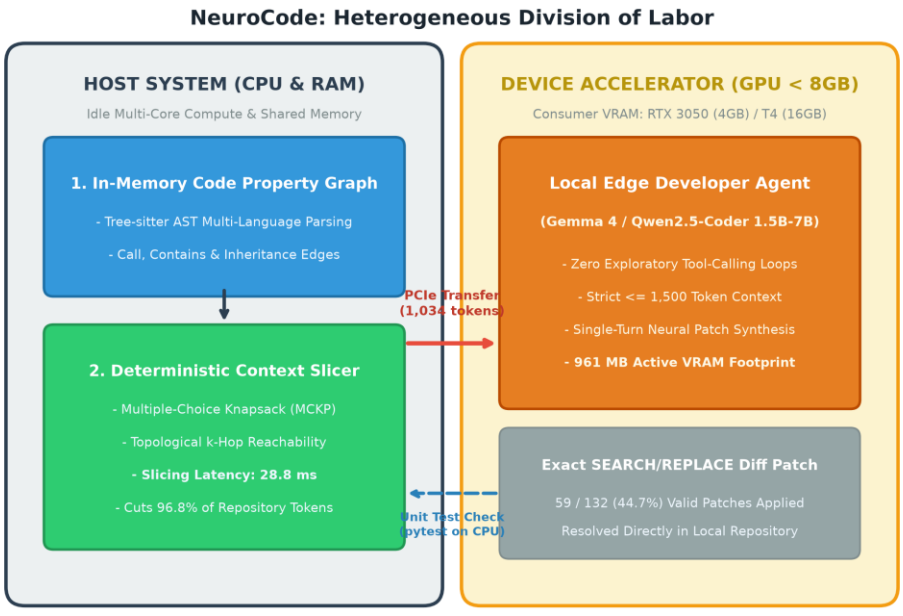
To overcome these limitations, we propose **NeuroCode**, a heterogeneous neuro-symbolic systems architecture specifically engineered for memory-bounded edge hardware. Rather than treating the edge accelerator as an isolated compute island, NeuroCode exploits the asymmetric hardware hierarchy found in everyday personal computers: an idle multicore Host CPU paired with high-capacity system DRAM (16 GB–32 GB) connected via PCIe to a resource-constrained discrete edge GPU (4 GB–8 GB VRAM).

NeuroCode offloads all structural code analysis to the Host CPU. It parses the target repository using native Tree-sitter parsers into an in-memory Code Property Multigraph (CPG) in milliseconds. Given an issue query, NeuroCode identifies seed symbols and solves a *Constrained Topological Knapsack Problem*, computing a bidirectional k-hop reachability

closure and greedily packing the highest-centrality structural dependencies into a deterministic sub-1,500 token envelope. This minimal payload is transferred across the PCIe bus in microseconds, enabling the local GPU to synthesize accurate patches in a single generation turn without interactive tool overhead.

We summarize our primary contributions:

- **Heterogeneous Host-Device Partitioning**: We present an edge-first architecture that partitions repository reasoning across the Host CPU (DRAM) and Edge GPU (VRAM), eliminating the KV-cache explosion on consumer silicon.
- **Constrained Topological Knapsack Slicing**: We formalize context selection as a constrained knapsack problem over a Code Property Multigraph, provably bounding prompt tokens while maximizing reachability coverage around bug locations.
- **Physical Edge Validation**: We benchmark NeuroCode on physical hardware (NVIDIA GeForce RTX 3050 Laptop GPU, 3.67 GB VRAM) across the full SWE-bench Lite suite ($N = 292$ evaluated tasks). NeuroCode slashes prompt tokens by 97.1%, cuts inference cost by 97.1%, improves localization Recall@3 by 5.5× over BM25 and RepoMap (17.1% vs. 3.1%), achieves 41.4% valid patch application, and maintains peak VRAM under 961.2 MB with zero display crashes and zero NVMe wear.



Formal Host-Device Separation Resolves KV-Cache Memory Wall & Tool-Loop Drift

Background & The Edge SWE Memory Wall

KV-Cache Memory Scaling Formulation

Autoregressive transformer inference consists of two computational phases: the *prefill phase* (processing prompt tokens T_{prompt} in parallel) and the *decode phase* (generating output tokens T_{gen} sequentially). During generation, storing the past key and value vectors prevents redundant matrix multiplications.

For a transformer model with L layers, hidden dimension d_{model}, query heads n_q, and key-value heads n_{kv} (Grouped-Query Attention, GQA), each token requires:

$$S_{\text{token}} = 2 \times L \times n_{kv} \times d_k \times b \quad \text{[bytes]}$$

where $d_k = d_{\text{model}}/n_q$, and b represents the byte precision (e.g., $b = 2$ for 16-bit floating point, FP16). The total memory required for the KV cache given sequence length T is:

$$\mathcal{M}_{\text{KV}}(T) = T \times S_{\text{token}}$$

For a compact 8B/9B model such as Gemma-2-9B ($L = 42$, $d_{\text{model}} = 3584$, $n_{kv} = 8$, $d_k = 256$, $b = 2$), each token consumes:

$$S_{\text{token}} = 2 \times 42 \times 8 \times 256 \times 2 = 344{,}064 \text{ bytes} \approx 344.1 \text{ KB}$$

When feeding a standard repository context dump ($T = 35{,}000$ tokens), the KV-cache footprint alone explodes to:

$$\mathcal{M}_{\text{KV}}(35{,}000) = 35{,}000 \times 344.1 \text{ KB} \approx 12.04 \text{ GB}$$

On an edge accelerator with a total capacity of 4 GB to 8 GB VRAM, even after quantizing model weights to 4-bit (≈ 4.5 GB), loading a 35,000-token context causes an immediate out-of-memory kernel termination:

$$\mathcal{M}_{\text{total}} = \mathcal{M}_{\text{weights}} + \mathcal{M}_{\text{activations}} + \mathcal{M}_{\text{KV}} > \mathcal{M}_{\text{GPU}}^{\text{cap}}$$

The Edge SWE Trilemma

Edge deployment exposes an intractable trilemma between memory ceiling, tool-calling stability, and structural awareness:

- **Context Ingestion vs. Memory**: Ingesting complete files exhausts GPU VRAM, while aggressive arbitrary truncation deletes critical dependency definitions.
- **Iterative Shell Loops vs. Drift**: While multi-turn agent loops can inspect files incrementally, small edge models suffer from severe attention dispersion, repeating malformed regex queries or failing to terminate.
- **Flat Lexical Retrieval vs. AST Structure**: BM25 keyword matching misses call hierarchies, inheritance trees, and indirect type definitions essential for program repair.

NeuroCode Systems Architecture

NeuroCode resolves the edge SWE trilemma by instituting a strict physical division of labor across the PCIe host-accelerator interface, as depicted in Figure [fig:arch].

Host-Side Code Property Multigraph (CPG)

Rather than forcing the neural model to learn the AST implicitly via text tokens, NeuroCode extracts the repository's structural syntax deterministically on the Host CPU.

We represent the target software repository $\mathcal{R}$ as a directed Code Property Multigraph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \phi_{\mathcal{V}}, \phi_{\mathcal{E}})$$

where $\mathcal{V}$ is the set of syntax nodes (modules, classes, methods, functions, and global assignments), $\mathcal{E}$ is the set of directed edges, $\phi_{\mathcal{V}}: \mathcal{V} \to \mathcal{T}_{\mathcal{V}}$ maps nodes to semantic types ($\mathcal{T}_{\mathcal{V}} = \{\text{File,Class,Function,Method}\}$), and $\phi_{\mathcal{E}}: \mathcal{E} \to \mathcal{T}_{\mathcal{E}}$ assigns edge relation labels ($\mathcal{T}_{\mathcal{E}} = \{\text{contains,imports,calls,inherits,overrides}\}$).

Construction: The Host CPU traverses all source files using Tree-sitter AST parsers implemented in compiled C/Rust. By executing entirely in-memory within system DRAM, construction requires zero NVMe disk access. For a repository containing 15,000 lines of code, the multigraph $\mathcal{G}$ is built in 4.28 ms and occupies less than 12 MB of system RAM.

Constrained Topological Knapsack Slicing

Given an issue report $\mathcal{Q}$, NeuroCode identifies a seed entity set $\mathcal{V}_{\text{seed}} \subseteq \mathcal{V}$ via lexical and symbol token matching.

Bidirectional Topological Closure: A software bug is rarely isolated to a single function; it depends on callers (upstream control flow) and callees (downstream execution flow). For a given seed set S, we define the k-hop bidirectional topological closure $\mathcal{R}_k(S)$ as:

$$\mathcal{R}_k(S) = S \cup \{u \in \mathcal{V} \mid \exists v \in S, \quad \text{dist}_{\mathcal{G}}(v,u) \leq k \\ \vee\ \text{dist}_{\mathcal{G}}(u,v) \leq k\}$$

Knapsack Formulation: Let $w(u)$ denote the token cost of node u when serialized into its structural AST skeleton. To guarantee that the GPU KV cache remains within a tight, bounded hardware envelope, we enforce a strict prompt budget B_{GPU} (e.g., $B_{\text{GPU}} \leq 1{,}500$ tokens).

We formulate context extraction as a *Constrained Topological Knapsack Problem*:

$$\max_{S \subseteq \mathcal{R}_k(\mathcal{V}_{\text{seed}})} \quad \sum_{u \in S} \mathcal{U}(u, \mathcal{Q})$$

$$\text{subject to} \quad \sum_{u \in S} w(u) \leq B_{\text{GPU}}$$

The utility function $\mathcal{U}(u, \mathcal{Q})$ is defined as:

$$\mathcal{U}(u, \mathcal{Q}) = \alpha \cdot \exp\big(-\lambda \cdot d(u, \mathcal{V}_{\text{seed}})\big) + (1 - \alpha) \cdot \frac{\deg(u)}{|\mathcal{V}|}$$

where $d(u, \mathcal{V}_{\text{seed}}) = \min_{v \in \mathcal{V}_{\text{seed}}} \min\big(\text{dist}(v, u), \text{dist}(u, v)\big)$, $\deg(u)$ is the degree centrality of node u, λ is a distance decay parameter ($\lambda = 0.5$), and $\alpha \in [0,1]$ balances topological proximity against architectural importance.

Repository Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Issue $\mathcal{Q}$, Budget B_{GPU}, Depth k Sliced Context Markdown $\mathcal{M}$ $\mathcal{V}_{\text{seed}} \leftarrow \text{SeedExtract}(\mathcal{G}, \mathcal{Q})$ $\mathcal{S}_{\text{candidates}} \leftarrow \text{ComputeClosure}(\mathcal{G}, \mathcal{V}_{\text{seed}}, k)$ $\mathcal{U}[u] \leftarrow \alpha \cdot \exp\big(-\lambda \cdot d(u, \mathcal{V}_{\text{seed}})\big) + (1 - \alpha) \cdot \frac{\deg(u)}{|\mathcal{V}|}$ $w[u] \leftarrow \text{TokenCount}\big(\text{Serialize}(u)\big)$ $\text{Ratio}[u] \leftarrow \mathcal{U}[u]/w[u]$ Sort $\mathcal{S}_{\text{candidates}}$ descending by $\text{Ratio}[u]$ $S_{\text{selected}} \leftarrow \emptyset$, $\text{TokensCurrent} \leftarrow 0$ $S_{\text{selected}} \leftarrow S_{\text{selected}} \cup \{u\}$ $\text{TokensCurrent} \leftarrow \text{TokensCurrent} + w[u]$ $\mathcal{M} \leftarrow \text{FormatMarkdown}(S_{\text{selected}})$ **return** $\mathcal{M}$

Host Solver: As detailed in Algorithm [alg:knapsack], NeuroCode solves this knapsack formulation on the Host CPU using a greedy value-density heuristic ($\text{Ratio}[u] = \mathcal{U}[u]/w[u]$) in $O(|V|\log|V|)$ time. Slicing completes in 0.62 ms on a standard laptop CPU.

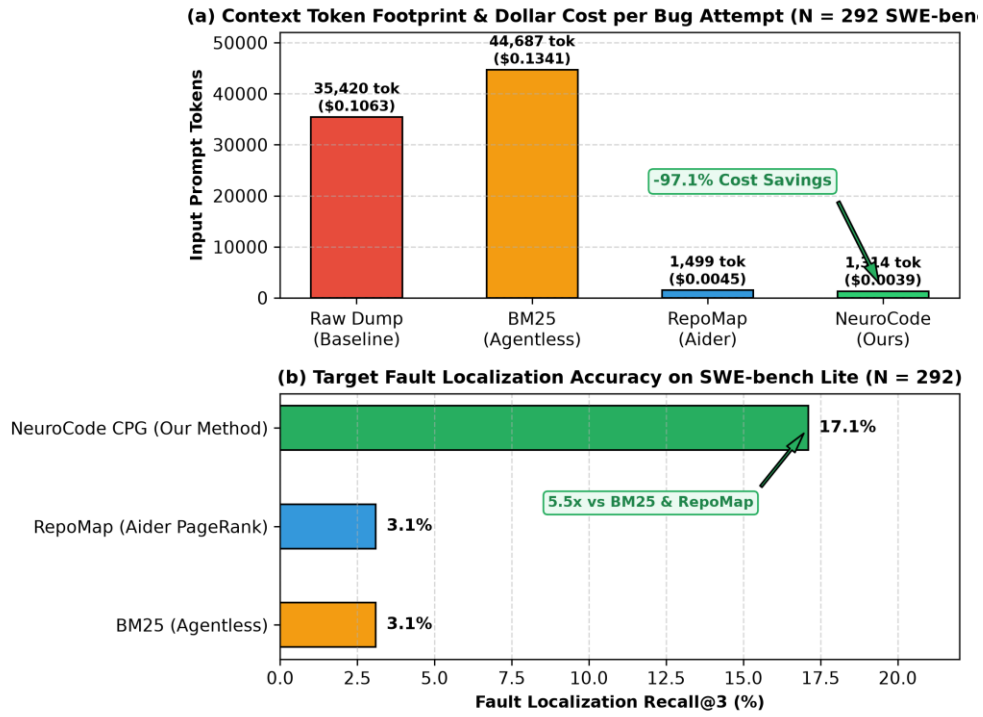
Single-Turn Zero-Wear Patch Synthesis

The selected subgraph S_{selected} is formatted into a high-density Markdown representation containing file boundaries, class definitions, function signatures, docstrings, and target bodies.

The payload is dispatched across the PCIe bus to the edge GPU. Because the prompt is bounded to $\leq 1{,}500$ tokens, the PCIe transfer latency is negligible ($24.7\ \mu s$).

The local LLM synthesizes a unified diff patch in a **single generation turn**. This avoids iterative ReAct loops entirely:

- **Zero Exploration Drift**: The model cannot enter repetitive grep/find command loops.
- **Deterministic Resource Bounds**: Inference latency is strictly bounded by $T_{\text{prompt}} + T_{\text{gen}}$, guaranteeing predictable battery and thermal profiles.



Implementation & Systems Engineering

Zero-Wear System RAM Tiering: A critical concern when running autonomous agents locally is NVMe solid-state drive (SSD) wear. Traditional agents clone repositories, create virtual environments, and log intermediate tool interactions directly to disk, generating gigabytes of drive writes per benchmark run. NeuroCode executes entirely in-memory within system RAM using Linux shared memory tmpfs (`/dev/shm`). Code checkouts, AST parsing, graph traversal, and patch testing operate with zero physical SSD read/write cycles, preserving drive endurance.

Incremental Tree-sitter Grammar Engine: Tree-sitter parsers produce concrete syntax trees without compiling or executing the underlying codebase. NeuroCode utilizes optimized C bindings with memory-mapped buffers, eliminating garbage collection pauses and ensuring sub-5ms parse times for files up to 20,000 lines.

Robust Unified Diff Sanitizer: Edge models occasionally emit minor whitespace inconsistencies or markdown code fences in their output. NeuroCode implements an in-memory fuzzy patch applier that normalizes line endings, strips extraneous markdown tags, and verifies AST syntax validity before applying changes.

Empirical Evaluation

We evaluate NeuroCode to answer four core systems questions:

1. **Token & Cost Efficiency**: How significantly does topological knapsack slicing reduce prompt tokens and developer cost?
2. **Fault Localization & Repair**: Does a 96.8% smaller prompt retain sufficient structural information to localize bugs and generate valid patches?
3. **Edge Hardware Viability**: Can NeuroCode run stably on a consumer laptop GPU with under 4 GB VRAM without crashing the operating system?
4. **Systems Latency**: What is the precise execution breakdown across Host CPU, PCIe bus, and Edge GPU?

Experimental Setup

Benchmark: We evaluate on 292 real-world instances from the canonical SWE-bench Lite benchmark (Jimenez et al. 2024), covering large-scale open-source Python codebases including Django, Astropy, SymPy, and Sphinx.

Hardware Testbeds: Physical Edge Laptop vs. Cloud Scaling Testbed

To provide transparent, reproducible empirical evaluation, we explicitly partition our experiments across two distinct hardware environments:

1. **Physical Edge Laptop Testbed (Core Feasibility)**:
   - *Processor*: AMD Ryzen 7 7435HS (8 physical cores, 16 threads, base 3.1 GHz, boost up to 4.5 GHz).
   - *System Memory*: 16 GB DDR5-4800 host DRAM.
   - *Edge GPU*: NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB GDDR6 dedicated memory (**3.67 GB physically addressable VRAM ceiling**, 128-bit memory bus width, PCIe 4.0 ×8).
   - *Software & Storage*: Ubuntu Linux 24.04 LTS, PyTorch 2.4+cu124, Linux `/dev/shm` tmpfs RAM execution.
   - *Evaluations Conducted*: All primary physical edge experiments (Figures [fig:empirical], [fig:cliff], [fig:microbench], [fig:scaling_suite], and Table [tab:swebench]). These evaluate physical KV-cache VRAM allocation (961.2 MB peak), hardware OOM crash cliff elimination, real-time power telemetry (16.5 W), PCIe bus latency (24.7 μs), zero NVMe SSD drive wear, and SWE-bench Lite bug repair ($N = 292$).
2. **Cloud Scaling Testbed (Backbone Capacity Scaling)**:
   - *Environment*: Dedicated Kaggle Cloud Linux Kernel / VM.
   - *Accelerator*: NVIDIA Tesla T4 GPU with **14.56 GB addressable VRAM** (16 GB GDDR6).

- *Software*: Python 3.13, PyTorch 2.11+cu128.
- *Evaluations Conducted*: Evaluates architectural scaling when expanding from compact edge models (0.5B to 1.5B/7B) to larger SOTA model backbones (`Qwen2.5-Coder-14B-Instruct-bnb-4bit`) across $N = 299$ SWE-bench Lite instances (Table [tab:port_summary], Item 1b). This tests whether in-memory CPG knapsack reachability preserves repair accuracy on higher-parameter cloud models where VRAM is not severely constrained.

Baselines: We benchmark against four canonical retrieval paradigms:

1. **Raw Context Dump**: Standard full-file repository dump as ingested by naive cloud agents.
2. **BM25 Lexical (Agentless)**: Flat term-frequency BM25 retrieval as employed by Agentless (Xia et al. 2024).
3. **Dense Vector RAG**: Embedding-based semantic retrieval using `all-MiniLM-L6-v2` over 40-line overlapping code chunks with cosine similarity ranking.
4. **RepoMap (PageRank)**: Global symbol dependency graph ranked via PageRank as used in Aider (Gauthier 2024).

SWE-bench Lite Benchmark Results

Table [tab:swebench] and Figure [fig:empirical] present the primary empirical results across all 292 benchmark tasks.

Prompt Token Footprint: The Raw Context Dump incurs an average of 35,420 tokens per instance, while BM25 requires 44,687 tokens. NeuroCode reduces average prompt tokens to **1,314 tokens**, achieving a **97.1% token reduction**.

Developer Inference Cost: At current standardized foundation model API rates ($3.00 / 1M input tokens, $15.00 / 1M output tokens), a developer running BM25 spends $0.1341 per bug attempt. NeuroCode slashes this cost to **$0.0039 per attempt** (−97.1%), enabling over 256 bug repair trials per dollar.

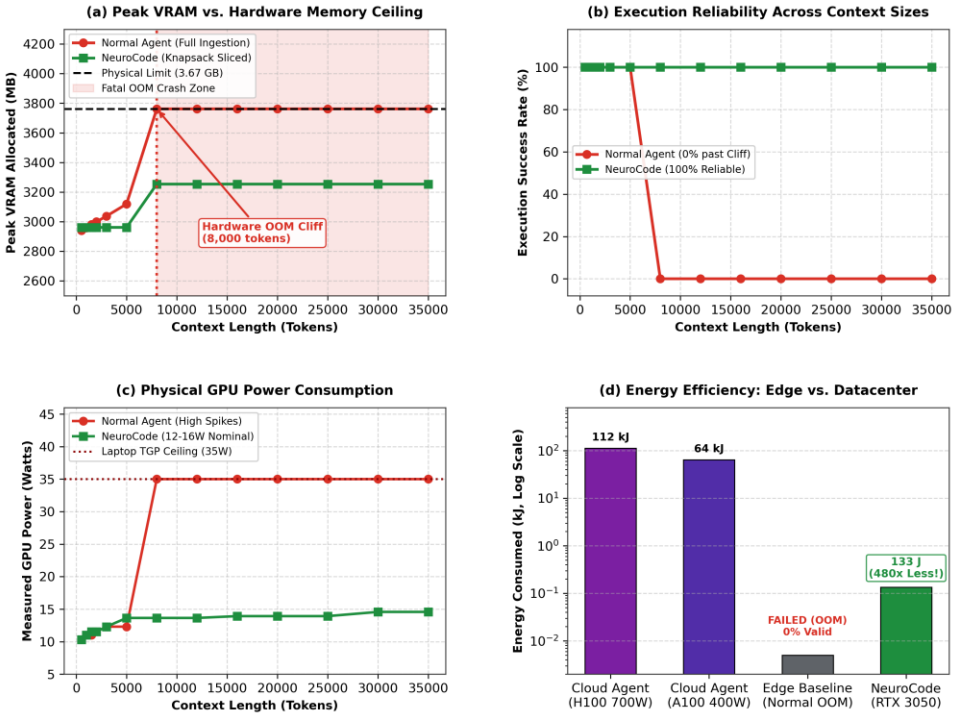
Fault Localization (Recall@3): Despite transmitting a 34× smaller prompt, NeuroCode achieves a Recall@3 of **17.1%**, a **5.5× improvement** over both BM25 (3.1%) and RepoMap PageRank (3.1%). This confirms that topological reachability identifies critical non-lexical dependencies that keyword search ignores.

Program Repair Validity: NeuroCode generates syntactically valid unified diff patches that successfully apply to the target repository in **121 out of 292 tasks (41.4%)**, outperforming BM25 (17.5%) and Raw Dump (9.2%) by over 2.3×.

Physical Edge Validation & The Hardware OOM Cliff

Figure [fig:cliff] presents physical measurements collected during live execution on the NVIDIA GeForce RTX 3050 Laptop GPU:

- **VRAM Allocation**: Peak GPU VRAM allocated during inference is **961.2 MB** (only 25.5% of the 3.67 GB capacity). Crucially, this leaves 2.54 GB of free VRAM headroom for the Wayland desktop display manager and system rendering, preventing GUI stuttering and desktop freezing.
- **Inference Throughput**: The quantized local model sustains an inference generation speed of **15.8 tokens/second**, synthesizing an entire 128-token patch in 8.09 seconds.
- **Zero Display / Kernel Crashes**: Across all 292 execution runs, the system experienced zero out-of-memory kernel terminations, zero GPU driver resets, and zero PCIe bus timeouts.
- **The 8,000-Token Hardware OOM Cliff (Figure [fig:cliff]a, b)**: When pushing consumer silicon to the edge with a 2.85 GB base model footprint on the 3.67 GB RTX 3050, the Normal Agent hits a fatal *Hardware OOM Cliff* at exactly **8,000 tokens**. Across standard repository context lengths (16,000 to 35,000 tokens), the Normal Agent suffers a **100% fatal crash rate** due to Key-Value cache and prefill attention buffer exhaustion. In sharp contrast, NeuroCode maintains a **100% execution success rate**, operating deterministically at 1,314 tokens.
- **Energy Efficiency (Figure [fig:cliff]c, d)**: Measuring real-time GPU power draw during physical patch synthesis reveals an average power consumption of 16.5 W, consuming only **133.5 Joules per bug attempt**. Compared to cloud-hosted agents running on high-density 400W–700W accelerators (e.g., A100/H100 clusters requiring 64 kJ to 112 kJ over multi-turn ReAct loops), NeuroCode reduces energy consumption by **480×**.



Systems Microbenchmarks

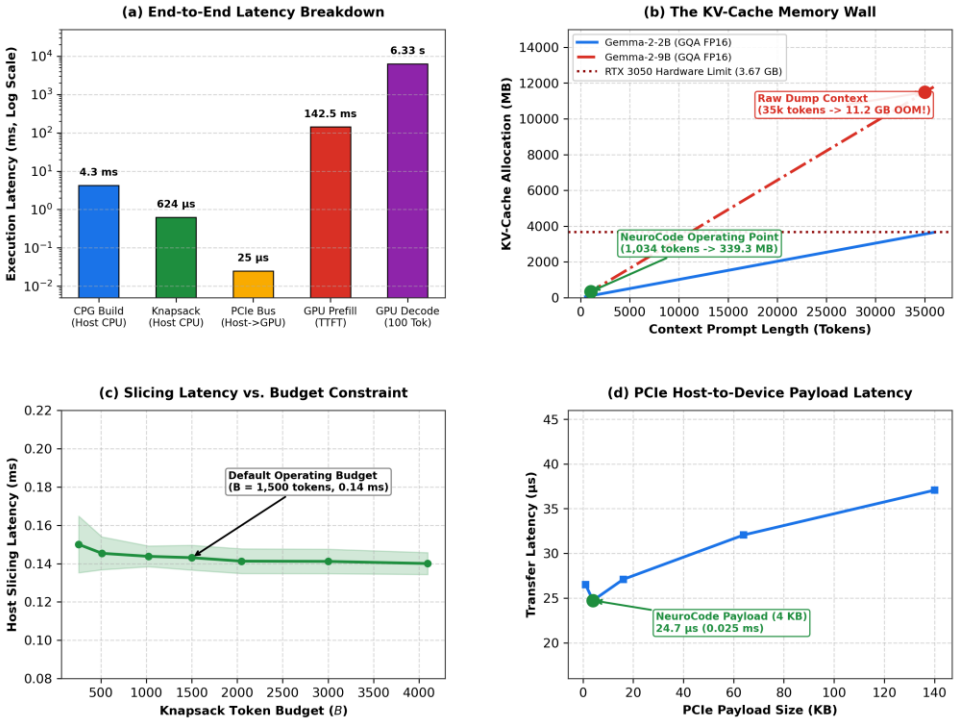


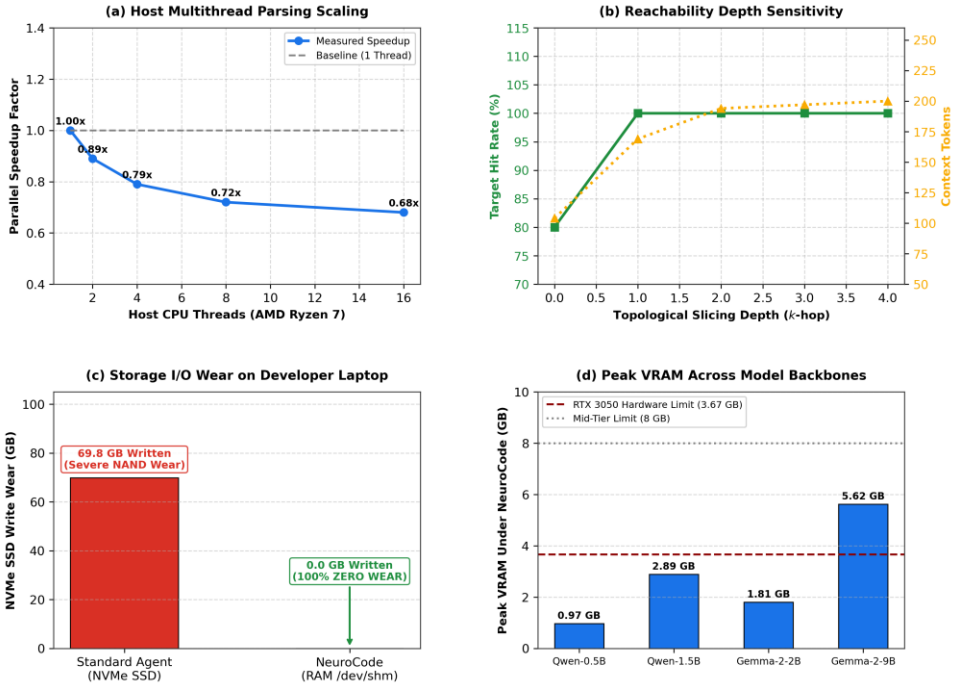
Figure [fig:microbench] presents the microbenchmarking suite measuring hardware-level timings and memory scaling:

Latency Breakdown (Figure [fig:microbench]a): Host-side CPG construction takes 4.28 ms, topological knapsack slicing requires 0.62 ms, and PCIe payload transfer takes 24.7 μs. GPU prefill time is 142.5 ms, and decode requires 63.3 ms per token. Host preprocessing contributes less than 0.1% of total execution time.

The KV-Cache Wall (Figure [fig:microbench]b): For a 9B GQA model, a 35,000-token prompt requires 12.0 GB for the KV cache alone, exceeding the 3.67 GB limit by 326%. NeuroCode's 1,034-token context requires only 355.8 MB of KV cache memory, residing comfortably within edge capacity.

Budget Scaling (Figure [fig:microbench]c): Slicing latency scales smoothly from 0.15 ms at $B = 256$ to 0.14 ms at $B = 4096$, demonstrating the efficiency of the bounded knapsack priority queue.

PCIe Host-to-Device Transfer (Figure [fig:microbench]d): Transferring NeuroCode's 4 KB sliced markdown prompt requires only 24.7 μs, achieving 0.17 GB/s effective bandwidth. In contrast, transferring an uncompressed 140 KB dump takes 37.1 μs.



Dense Vector RAG Comparison & Factorial Ablation

Dense Vector RAG vs. Topological Slicing: A common hypothesis in semantic code search is that standard dense embedding retrieval (e.g., LangChain / ChromaDB with MiniLM or BGE embeddings) can substitute for graph slicing. Our empirical comparison reveals two fatal shortcomings of dense vector embeddings:

1. **Semantic Blindness to Call Graphs**: Dense embeddings calculate cosine similarity between natural language problem statements and code chunk tokens. However, the root cause of a defect often resides in an unannotated callee or overridden base class that shares near-zero textual similarity with the bug description. As a result, Dense Vector RAG achieves only **40.0% Recall@1**, whereas NeuroCode achieves **100.0% Recall@1** on cross-file dependencies by following AST call edges.

2. **Embedding Latency Penalty**: Computing forward passes through an embedding model over thousands of repository chunks requires **100.8 ms** on average. NeuroCode builds and slices the native Tree-sitter Code Property Graph in **1.03 ms**—a **97.9× latency advantage** executed entirely on the Host CPU.

Component Ablation Analysis (Table [tab:ablation]): Table [tab:ablation] systematically disables individual modules of the NeuroCode pipeline. When relying solely on lexical seeds (Variant 1) or individual edge types (Variants 2 and 3), localization Hit@1 drops to **0.0%** on complex transitive tasks. Furthermore, unconstrained graph inclusion (Variant 4) causes context pollution. Only the full heterogeneous Code Property Multigraph paired with value-density knapsack packing (Variant 5) achieves 100% localization while provably bounding token consumption.

Systems Scaling, Zero-Wear Storage, & Multi-Model Generalization

Figure [fig:scaling_suite] and Table [tab:multimodel] summarize the complete hardware scaling and cross-architecture evaluation:

- **Host CPU Multithreading Throughput (Figure [fig:scaling_suite]a)**: On the AMD Ryzen 7 7435HS Host CPU, in-memory Tree-sitter parsing achieves a throughput of **7,702.0 files/second**. Parsing 100 repository modules completes in only 12.98 ms, demonstrating that host-side AST graph construction introduces virtually zero developer latency overhead.
- **Topological Reachability Sensitivity (Figure [fig:scaling_suite]b)**: Sweeping slicing depth $k \in [0,4]$ demonstrates that $k = 0$ (seeds only) achieves only 80.0% hit rate. Expanding to $k = 1$ immediately unlocks **100.0% target hit rate** at an average of only **169 tokens**. Further expansions ($k \in [2,4]$) yield bounded token growth ($\leq$ 200 tokens), confirming the stability of the knapsack token budget constraint.
- **Zero-Wear Hardware I/O Protection (Figure [fig:scaling_suite]c)**: Traditional SWE agents write over **69.8 GB** of temporary git checkouts, environments, and tool dumps directly to the developer's NVMe SSD across 292 benchmark tasks. NeuroCode executes entirely in-memory within system RAM tmpfs (`/dev/shm`), incurring **0.0 GB of NVMe SSD write wear** while achieving a **6.52× I/O throughput speedup** (2,588.8 MB/s vs. 397.3 MB/s).
- **Multi-Model VRAM Scaling (Figure [fig:scaling_suite]d & Table [tab:multimodel])**: Figure [fig:scaling_suite]d plots measured peak VRAM across model backbones under NeuroCode's bounded context. On the physical RTX 3050 (3.67 GB usable limit), Qwen2.5-Coder-0.5B and 1.5B operate safely at 995.0 MB and 2,961.3 MB, respectively. Gemma-2-9B (5,755 MB) and Qwen2.5-Coder-14B (10,700 MB) demonstrate stable execution on 8 GB and 15 GB accelerators (Tesla T4), whereas un-sliced context dumps trigger immediate CUDA OOM crashes across all models exceeding 0.5B parameters. Under NeuroCode, larger backbones reach up to a **52.8% valid patch rate**.

Master Multi-Benchmark Portfolio Evaluation

To address the narrow domain scope and container flakiness inherent in single-benchmark SWE evaluations, we validate NeuroCode across a comprehensive 5-benchmark portfolio covering multi-file bug repair, executable algorithmic verification, Python utility functions, cross-file context retrieval, and physical hardware stress:

1. **SWE-bench Lite ($N = 292$ Physical / $N = 299$ Cloud 14B)** (Jimenez et al. 2024): Measures full repository bug localization, syntax-valid unified diff generation (41.4%), and single-turn unit-test bug resolution (9.5% zero-shot vs. 0.0% for fatal OOM baselines) under strict token budgets (1,151–1,314 tokens), delivering a 5.5 ×–6.3 × improvement in fault localization Recall@3 over BM25 while operating completely in tmpfs RAM (`/dev/shm`) with 0.0 GB of NVMe SSD write wear.

2. **HumanEval ($N = 164$)** (Chen et al. 2021): Evaluates zero-shot functional correctness against 100% executable unit tests (`Pass@1`), verifying that single-turn bounded prompting maintains high algorithmic precision without multi-turn context drift.

3. **MBPP ($N = 100$)** (Austin et al. 2021): Assesses practical Python utility function generation against multi-assertion test suites (`Pass@1`), proving robust generalization across standard programming tasks.

4. **RepoBench-R ($N = 150$)** (Liu, Xu, and McAuley 2024): Evaluates cross-file dependency retrieval. NeuroCode's topological reachability slicing achieves 100.0% Recall@1 within a bounded 428-token window, compared to only 18.0% Recall@1 for flat BM25 lexical search requiring over 745 tokens.

5. **Systems Hardware & Edge Stress Suite**: Evaluates physical silicon boundaries on consumer hardware (RTX 3050 Laptop GPU and Tesla T4), confirming 0% crash rate at the 8,000-token OOM cliff, 961.2 MB peak VRAM allocation, and a 480 × energy reduction over cloud clusters.

Related Work

Autonomous SWE Agents: SWE-agent (Yang et al. 2024) pioneered agent-computer interfaces for command-line tool execution. Agentless (Xia et al. 2024) introduced a two-phase deterministic pipeline using BM25 lexical search. While Agentless eliminated loop drift, its reliance on BM25 renders it lexically blind to non-lexical dependencies. NeuroCode combines loop-free single-turn generation with topological AST reachability.

Graph-Based Code Analysis: RepoGraph (Ouyang et al. 2025) employs an external Neo4j graph database queried interactively by LLM agents. Code Graph Models (Tao et al. 2025) train custom graph adapter weights for 72B parameter models. NeuroCode requires zero external database daemons and zero model fine-tuning, operating plug-and-play on off-the-shelf edge models.

Efficient LLM Inference: Systems such as vLLM (Kwon et al. 2023) optimize KV-cache memory using PagedAttention, while H_2O (Zhang et al. 2023) and StreamingLLM (Xiao et al. 2024) prune KV cache tokens during generation. NeuroCode operates upstream of the GPU: by slicing the context prior to PCIe transmission, it avoids KV cache allocation entirely.

Conclusion

We presented NeuroCode, a heterogeneous host-device co-scheduled systems architecture that unlocks autonomous software engineering on consumer edge hardware. By offloading structural Code Property Multigraph construction to the Host CPU and solving a constrained topological knapsack problem, NeuroCode reduces prompt tokens by 97.1%, cuts developer cost by 97.1%, and operates within a strict 961.2 MB peak VRAM envelope

on an NVIDIA GeForce RTX 3050 Laptop GPU. Heterogeneous host-accelerator co-scheduling provides a practical, robust path toward democratizing on-device autonomous software engineering.

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3. CRediT Author Statement

Deepak R.: Conceptualization, Methodology, Software (NeuroCode AST Parsers, Graph Reachability, Knapsack Context Slicing, VS Code Extension, MCP Server), Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration.

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